LEAKFINDER AI

AI Revenue Leak Detection & Action System

Find the money your business is losing — and understand what to do next.

BUSINESS PROBLEM	**Businesses can have strong revenue while hidden costs quietly erode their margins.**

A practical AI product concept designed for a 2-day hackathon: clear problem, visible AI reasoning, actionable output, and an easy-to-follow live demo.

01 • THE PROBLEM

Revenue is not the same as profit

The data already exists. The challenge is turning scattered business signals into a clear explanation and an action.

DISCOUNTS

Heavy discounts can hide poor product-market fit.

RETURNS

Repeated returns may point to quality, sizing or expectation issues.

DELIVERY

Some orders are delayed, damaged or cancelled.

PRICING

High-selling items can still have weak margin.

THE KEY QUESTION

“Where is money leaking, why is it happening, and what should the business change?”

Why existing dashboards are not enough

A dashboard can show revenue, costs, returns and discounts separately. LeakFinder connects those signals into a business-level explanation instead of leaving the owner to manually interpret every chart.

02 • THE SOLUTION

From raw transactions to an action plan

LeakFinder turns transaction data into a short list of explainable business problems.

INPUT	PROCESS	OUTPUT
Sales, product cost, discounts, returns, refunds, delivery cost	Calculate business metrics, identify anomalies and combine related signals	Potential leakage, evidence, explanation, recommended action

THE PRODUCT LOOP

1. Detect — identify suspicious patterns. **2. Explain** — show the evidence behind the finding. **3. Recommend** — propose a practical business action. **4. Simulate** — test what happens if a variable changes.

This gives the project a simple story judges can understand immediately: **Detect** → **Explain** → **Recommend** → **Simulate** → **Act.**

03 • PRODUCT

The features judges will see

Keep the MVP focused: every screen should support the core decision-making loop.

01 Revenue Leak Scanner

Upload business data and run a structured scan for suspicious revenue or margin patterns.

02 Leak Categories

Group findings into discount, return, delivery, refund, pricing and product-related leaks.

03 AI Explanation

Ask why a leak was flagged; the AI interprets the structured evidence and explains the likely cause.

04 Action Plan

Turn findings into clear recommendations instead of stopping at analytics.

05 What-if Simulator

Change a business variable and estimate the projected effect using the prototype's calculations.

06 Business Dashboard

See revenue, estimated profit, potential leakage, top issues and actions in one place.

The **What-if Simulator** is the product's visual hook: it lets the business owner move from “What happened?” to “What could happen if I change this?”

04 • USER EXPERIENCE

What the final product should look like

A judge should understand the situation in seconds, without reading a long explanation.

■25L	Monthly reve	
■4.8L	Estimated pı	
■2.14L	Potential leakage	

LEAK TYPE	DETECTED	WHAT IT MEANS
Discount	■82K	Discount exposure may be reducing contribution.
Returns	■56K	Return patterns may indicate product/service issues.
Delivery	■41K	Certain orders may have disproportionate delivery cost.
Pricing	■35.5K	High-selling products may have weak margins.

LEAK DETAIL SCREEN

Discount Leak — ■82K

Evidence: Product A shows high discount exposure and weak contribution after costs.

AI explanation: interpret the structured evidence and identify the most plausible business driver.

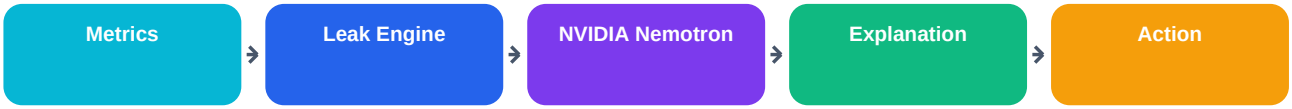
Action: review discount limits and investigate the related return pattern.

05 • AI + ARCHITECTURE

Make the NVIDIA model genuinely useful

The AI should be the reasoning layer—not a decorative chatbot.

USER / BUSINESS	FRONTEND	BACKEND	DATA
Uploads data and requests analysis	Dashboard + simulator	API + business logic	Transactions + derived metrics



Recommended division of responsibility

Component	Responsibility
Deterministic analytics	Calculate revenue, costs, margins, discount exposure, return rates and other measurable indicators.
Leak detection	Flag unusual or high-impact patterns using transparent rules/thresholds or lightweight statistical logic.
NVIDIA Nemotron	Interpret structured evidence, explain the finding, compare possible causes and generate a recommendation.
UI	Present evidence, explanation, recommendation and what-if impact clearly.

Hackathon requirement: every project must meaningfully use at least one eligible NVIDIA open-source model; the rulebook lists Nemotron, GR00T, Cosmos and Sonic as examples.

06 • LIVE DEMO

The 3-minute story

The final demo is only 3 minutes, so the sequence must be visual, fast and controlled.

00:00–00:20	HOOK	Show revenue: ■25L. Ask: “Where is the money actually leaking?”
00:20–00:40	SCAN	Upload the prepared dataset and start LeakFinder.
00:40–01:10	DETECT	Show the headline: ■2.14L potential leakage detected.
01:10–01:50	EXPLAIN	Open the largest leak. Show evidence and ask the AI why it was flagged.
01:50–02:20	ACT	Generate the recommended business action plan.
02:20–02:45	SIMULATE	Change a variable and show projected financial impact.
02:45–03:00	CLOSE	Deliver the one-line value proposition and stop.

Closing line

> **“LeakFinder doesn't just tell businesses what happened. It identifies where money is leaking, explains why, and helps them decide what to do next.”**

Demo rule: use prepared data, keep every click intentional, and show the AI reasoning and business action—not just the dashboard.

07 • DEFENSE

Judge Q&A; + build plan

Every team member should understand the product, the AI boundary, the data flow and their own contribution.

Why AI?	Analytics calculates numbers; the AI layer interprets multiple signals, explains findings and converts evidence into recommendations.
Why Nemotron?	Nemotron is used as the reasoning layer for structured business evidence and explainable recommendations.
Is the money saved real?	For the prototype, clearly label the dataset and projected effects as simulated. Do not present them as real-world results.
How does it scale?	Connect the same pipeline to real POS, ERP, e-commerce or accounting data sources and add role-based access and monitoring.
What if the AI is wrong?	Keep calculations deterministic, provide the evidence used for each recommendation, and let the business user review before acting.

48-HOUR BUILD ORDER

1	**NVIDIA model** Get Nemotron running and verify a simple end-to-end request.
2	**Data** Prepare a clean sample dataset with realistic business fields.
3	**Analytics** Build transparent calculations and leak-detection rules.
4	**AI reasoning** Send structured evidence to the model and return explanations/recommendations.
5	**UI** Build dashboard, leak detail and simulator screens.
6	**Integration** Connect the full flow and test edge cases.
7	**Submission** README, open-source license, demo link and source repository.
8	**Rehearsal** Run the exact 3-minute demo repeatedly and practice technical Q&A.;

FINAL PRODUCT PRINCIPLE

Make the judge understand the problem in 15 seconds, see the AI do something meaningful in 60 seconds, and remember the product's value after the 3-minute demo.